

PRACHI HEMANT DUDHE

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SUMMARY

Data Scientist with hands-on experience designing experiments, analyzing results under uncertainty, and translating statistical findings into clear decisions. Built and evaluated AI systems with benchmarked quality metrics, applied hypothesis-test-iterate cycles to diagnose and fix model failures, and communicated trade-offs clearly to non-technical stakeholders.

EDUCATION

Texas A&M University, College Station, TX

Master of Science in Data Science (ECE Track) — GPA: 3.5/4.0

Aug 2024 to May 2026

College Station, TX

Pune University

Bachelor of Engineering in Electronics and Telecommunication, Honors in Data Science — GPA: 3.8/4.0

Aug 2018 to May 2022

India

TECHNICAL SKILLS

- **Experimentation & Statistics:** A/B testing, controlled experiments, hypothesis testing, effect size, variance analysis, anomaly detection, statistical EDA, metric selection, causal reasoning
- **ML & Decision Science:** scikit-learn, PyTorch, regression, classification, reinforcement learning, model evaluation, feature engineering
- **AI-Enabled Analytics:** RAG pipelines, LlamaIndex, FAISS, semantic embeddings, LLM integration, retrieval quality benchmarking
- **Data Engineering:** Python (Pandas, NumPy), SQL, ETL pipelines, schema validation, data quality monitoring, Git

PROFESSIONAL EXPERIENCE

Research Data Engineer

Texas A&M University - Corpus Christi

Aug 2025 to Dec 2025

Texas

- Agricultural trial data was siloed across 29 inconsistent PDFs with no standardized schema, making variety comparisons impossible - Built a 3-stage data normalization pipeline for agricultural trial data, reducing malformed records by 30%
- With clean data in place, designed and benchmarked a controlled evaluation framework for a hybrid query system, comparing SQL and semantic RAG approaches across query types to identify the most reliable method for each use case
- Translated system outputs into actionable recommendations for 3 cross-functional research teams via an interactive Streamlit dashboard with 3 query modes, communicating trade-offs and uncertainty clearly so teams could drive weekly operational decisions rather than just consume reports

Data Science Intern

Sulzer

Jun 2025 to Jul 2025

India

- Employees were spending hours manually searching through technical PDFs to answer routine operational questions - built a prototype RAG pipeline using PyPDF2 to chunk 500+ document segments, generate all-MiniLM-L6-v2 embeddings (384-dim), and index them in FAISS for semantic retrieval
- Benchmarked retrieval performance across embedding configurations and chunk sizes to find the optimal trade-off between precision and latency, achieving 20-40 ms query response on CPU and documented methodology, assumptions, and a scalability roadmap for enterprise deployment

Data Analyst

Tata Consultancy Services

Oct 2022 to Jul 2024

India

- Production and supply chain teams had no reliable, automated view of operational health - built end-to-end ETL pipelines extracting and transforming enterprise ERP data, implementing statistical anomaly detection to flag data inconsistencies before they distorted KPIs, and delivered Power BI and Tableau dashboards tracking 10+ metrics adopted by management
- Optimized complex SQL queries by restructuring joins and indexing, reducing average pipeline runtime by 25%
- Partnered with Operations and Sales to define decision-critical metrics, resolve cross-system discrepancies, and create a trusted reporting foundation for faster operational and business decisions

TECHNICAL PROJECTS

Acing Blackjack - Controlled Experimentation and Model Evaluation | Python, PyTorch, Double DQN

- Standard reward shaping was producing a 10-point performance gap vs. optimal strategy - diagnosed the failure through variance and convergence analysis (32-35% vs. target 42-44%), identified reward hacking as root cause, redesigned to sparse terminal rewards, and validated full recovery to 43.2%
- Designed and analyzed controlled experiments across learning rate, replay buffer, and update frequency to quantify their effect on policy stability and performance across 400,000+ training episodes
- Built a staged training curriculum to reduce catastrophic forgetting and improve transfer across training phases

Road Lane Overlay - Model Comparison and Evaluation | Python, PyTorch, CLRRerNet, Qwen-Image-Edit, LoRA

- Improved lane reconstruction performance by testing a semantically guided editing approach, increasing F1 by 7.1% and improving FID by 12.3%
- Fine-tuned Qwen-Image-Edit with LoRA adapters to restore missing lane markings using control overlays for geometric guidance
- Processed 62,532 road images with CLRRerNet to generate lane masks and limit edits only to missing-lane regions

Fitbit Health Analytics - Statistical Analysis and Insight Generation | Python, Pandas, SQL, Plotly

- Conducted statistical analysis and controlled comparisons on 10,000+ time-series health records to identify behavioral patterns and evaluate changes across key metrics
- Delivered interactive dashboard framing findings as decisions with uncertainty acknowledged - designed for both technical analysts and non-technical stakeholders to explore results independently